

DSC 640 Week 3&4_White_House_Logs -Ethical Analysis and outcomes

January 20, 2026

1 White House Visitor Access Records: Data Ethics Analysis

1.1 A Story of Transparency, Privacy, and Public Interest (2021-2023)

1.2 Executive Summary

This analysis examines 1,207,458 visitor access records from the White House Worker and Visitor Entry System (WAVES) spanning from 2021 to 2023. This dataset represents a critical intersection of government transparency, personal privacy, and public accountability. Through careful analysis and ethical consideration, we present findings that illuminate both the promise and peril of open government data.

Call to Action: As data practitioners, researchers, and citizens, we must demand transparent data practices while protecting individual privacy rights. This analysis serves as a framework for ethical data handling in government transparency initiatives.

1.3 1. Data Transformation and Changes Made

1.3.1 1.1 Initial Data State

The original dataset consisted of 25 separate CSV files: - **2021 records:** 41,412 entries (1 file) - **2022 records:** 363,830 entries (12 monthly files) - **2023 records:** 802,216 entries (12 monthly files)

1.3.2 1.2 Transformations Applied

Consolidation

- **Action:** Combined 25 individual CSV files into a single unified dataset
- **Justification:** Enable longitudinal analysis and pattern detection across the entire administration period
- **Impact:** Created a single dataset of 1,207,458 records

Column Standardization Issues Identified The analysis revealed **inconsistent column naming conventions** across files: - 2021 file used: NAMELAST, NAMEFIRST, ACCESS_TYPE, etc. (UPPERCASE) - 2022-2023 files used: Last Name, First Name, Access Type, etc. (Title Case)

Result: The combined dataset contains **duplicate columns** (50 columns instead of ~27 expected)

CRITICAL ASSUMPTION: We retained both column sets WITHOUT merging them, creating a dataset where: - ~93% of records have data in Title Case columns - ~7% of records (2021 only) have data in UPPERCASE columns

Metadata Addition

- **Action:** Added `source_file` column to track record origin
- **Justification:** Maintain data provenance and enable source-level filtering
- **Impact:** 100% traceability of each record to its original file

Data Type Preservation

- **Action:** Loaded all data with minimal type conversion
- **Justification:** Preserve original data fidelity; avoid information loss
- **Impact:** Dates remain as strings; further parsing required for temporal analysis

1.3.3 1.3 What Was NOT Changed

- No records were deleted or filtered
- No personal information was redacted beyond original release
- No values were modified or “cleaned”
- No imputation of missing values was performed

1.4 2. Legal and Regulatory Guidelines

1.4.1 2.1 Applicable Legal Framework

Presidential Records Act (PRA) - 44 U.S.C. §§ 2201-2209

- Governs creation, management, and preservation of Presidential records
- Requires systematic documentation of White House activities
- WAVES records fall under this mandate

Freedom of Information Act (FOIA) - 5 U.S.C. § 552

- Provides public access to federal agency records
- White House voluntarily discloses WAVES records (not legally required)
- Balances transparency with privacy exemptions (b)(6) and (b)(7)(C)

Privacy Act of 1974 - 5 U.S.C. § 552a

- Protects personal information in government systems
- Requires notice of records systems and usage
- Limits disclosure of personally identifiable information (PII)

E-Government Act of 2002

- Section 208 requires Privacy Impact Assessments (PIAs)
- Mandates privacy protections for federal information systems

1.4.2 2.2 White House Policy Commitments

The Obama, Trump, and Biden administrations have each committed to varying degrees of WAVES disclosure: - **Obama Administration (2009-2017)**: Pioneered voluntary monthly WAVES releases - **Trump Administration (2017-2021)**: Discontinued regular releases; released limited records after administration - **Biden Administration (2021-present)**: Resumed quarterly releases with 90-day delay

1.4.3 2.3 Compliance Status

Data was legally obtained from official White House WAVES releases

Delayed release (90-120 days) balances security concerns with transparency

Voluntary disclosure - no legal mandate requires these releases

1.5 3. Risk Assessment: Transformations and Visualizations

1.5.1 3.1 Privacy Risks

Re-identification Risk HIGH Risk: Combining visitor names with publicly available data could enable: - Cross-referencing with social media profiles - Tracking individual visit patterns over time - Inferring relationships between visitors and officials - Revealing sensitive meetings (e.g., medical consultations, legal consultations)

Example Scenario:

Record: John Smith visited "Visitee: Medical Unit" on 15 occasions

Public risk: Could reveal health information about an administration official

Location Tracking MEDIUM Risk: Meeting locations and times create movement patterns: - 929,446 meetings at White House main building - 162,538 meetings at OEOP (Eisenhower Executive Office Building) - Temporal patterns could reveal security vulnerabilities

Association Inference MEDIUM Risk: Visitor patterns could reveal: - Undisclosed business relationships - Policy influence networks - Early warning of policy initiatives (e.g., industry leaders visiting before announcements)

1.5.2 3.2 Visualization Risks

Context Collapse Visualizations that show “most frequent visitors” without context could: - Misrepresent routine White House tour participants as “influential” - Conflate public tours with private policy meetings - Create misleading narratives about access and influence

Temporal Distortion Time-series visualizations could: - Highlight unusual visit patterns that have innocuous explanations - Draw attention to “gaps” that reflect security protocols or personal privacy

Aggregation Bias Summary statistics like “Top 10 Visitees” include: - 676,351 records with “Visitee: Office, Visitors” (likely public tours) - This **overwhelms** actual policy-relevant visits in aggregate visualizations

1.5.3 3.3 Security Risks

Pattern Recognition HIGH Adversaries could analyze: - When high-value targets (e.g., foreign dignitaries) visit - Security checkpoint usage patterns (POA/POD fields) - Staffing patterns based on “Last Updated By” data

Social Engineering MEDIUM Bad actors could use this data to: - Impersonate legitimate visitors - Research staff names for phishing attacks - Identify optimal times for facility access attempts

1.6 4. Assumptions and Data Filtering Transparency

1.6.1 4.1 Explicit Assumptions Made

Assumption 1: Column Name Inconsistency

- **Assumption:** Title Case and UPPERCASE columns represent the same data fields
- **Risk:** May have merged different data types if assumption is incorrect
- **Mitigation:** Source file tracking allows reversal

Assumption 2: “VA” Access Type Uniformity

- **Observation:** 99.99%+ of records have Access Type = “VA” (Visitor Appointment)
- **Assumption:** This represents standard visitor access protocol
- **Unknown:** What other access types exist? Are they excluded?

Assumption 3: Missing Data is Intentional

- **Observation:** 91% of TOA (Time of Arrival) fields are empty
- **Assumption:** Records are created at appointment scheduling, not check-in
- **Alternative:** Data may be redacted post-hoc for security

Assumption 4: Public Tours vs. Official Meetings

- **Observation:** 676,351 records (56%) have “Visitee: Office, Visitors”
- **Assumption:** These are public tours, not policy meetings
- **Risk:** Could include actual visits misclassified in entry system

1.6.2 4.2 Data Filtering Disclosure

NO FILTERING WAS APPLIED IN THIS ANALYSIS We explicitly chose to retain all records, including: - Records with missing values (some columns 93% incomplete) - Duplicate-appearing records (same visitor, same day, different UINs) - Records with minimal information (name + date only)

Rationale for No Filtering:

1. **Transparency:** Any filtering decision introduces bias
2. **Preservation:** Missing data may be informative (e.g., security redactions)
3. **User Choice:** Downstream analysts can apply their own filters with full knowledge

1.6.3 4.3 Labeling of Missing Data

Column	Missing %	Likely Reason
TOA (Time of Arrival)	70%	Appointments vs. actual arrivals
POD (Point of Departure)	90%	Not tracked systematically
APPT_CANCEL_DATE	99.96%	Most visits are NOT cancelled
BDGNBR (Badge Number)	86%	Only issued for approved access
CALLER_ROOM	100%	Column may be deprecated

Documentation Practice: All missing data percentages documented in analysis output

1.7 5. Data Sourcing and Credibility Verification

1.7.1 5.1 Primary Source

Origin: Official White House WAVES releases

Publisher: Executive Office of the President

Distribution: WhiteHouse.gov and Archive.gov

Release Schedule: Quarterly with 90-day delay

1.7.2 5.2 Verification Methods Applied

Cross-Reference Verification

- Compared record counts against official White House announcements
- Verified file naming conventions match official releases
- Confirmed release dates align with historical White House disclosure schedule

Temporal Consistency Checks

- Verified chronological progression (2021 → 2023)
- Confirmed no records from “future dates” relative to release dates
- Found records with APPT_MADE_DATE after APPT_START_DATE (logically inconsistent)

Statistical Plausibility

- Visitor counts align with reported White House foot traffic
- Peak periods correlate with known public events (e.g., December 2022: 87,061 records likely includes holiday tours)
- Some records show TOTAL_PEOPLE = 1000+ (may indicate group tours or system codes)

1.7.3 5.3 Known Data Quality Issues

1. **Inconsistent Column Naming** (as detailed in Section 1.2)
2. **Duplicate Records Possible:** Same visitor, same date, different UIN codes
3. **Incomplete Temporal Data:** Arrival/departure times often missing
4. **Name Variations:** “Office, Visitors” vs. “Visitors Office” vs. “Visitors, Office”

1.7.4 5.4 Credibility Assessment

Criterion	Rating	Justification
Authenticity		Official government source
Completeness		Many fields sparsely populated
Accuracy		Likely accurate but unverifiable
Consistency		Column naming issues, format changes
Timeliness		90-day delay standard for security

1.8 6. Ethical Acquisition and Usage

1.8.1 6.1 Was the Data Acquired Ethically? YES

Consent and Notice

- White House visitors are informed their visit information may be publicly released
- Signage and forms notify visitors of WAVES policy
- **However:** Extent of awareness varies (casual tour-goers vs. official visitors)

Legitimate Purpose

- Government transparency is a **core democratic value**
- Public has legitimate interest in White House access patterns
- Balances against individual privacy through delayed release

Legal Authorization

- White House has explicit authority to release this data
- No hacking, breach, or unauthorized access occurred
- Data obtained through official channels

1.8.2 6.2 Ethical Concerns Despite Legal Acquisition

Concern 1: Granularity of Personal Data

- **Issue:** Names + dates + locations enables tracking individuals
- **Question:** Is this level of detail necessary for transparency goals?
- **Alternative:** Could aggregate statistics serve same purpose?

Concern 2: Differential Privacy Impact

- **Issue:** High-profile visitors vs. average citizens
- **Public figures:** Expect reduced privacy; visits are newsworthy
- **Average citizens:** May not anticipate persistent public record
- **Example:** A constituent meeting their representative shouldn't become a permanent internet record

Concern 3: Secondary Use

- **Issue:** Data released for transparency, but usable for:
 - Opposition research
 - Stalking/harassment
 - Marketing/targeting
- **Ethical question:** Should data use be restricted post-release?

Concern 4: Perpetual Accessibility

- **Issue:** Internet archives preserve this data indefinitely
 - **Right to be forgotten:** Not applicable in US law
 - **Impact:** A 2021 visitor’s data remains searchable in 2026, 2036, 2046...
-

1.9 7. Ethical Risk Mitigation Strategies

1.9.1 7.1 Technical Safeguards

Implemented in This Analysis:

1. **No Unnecessary Aggregation of Sensitive Fields**
 - Avoided creating “visitor profiles” linking all visits per person
 - No social network graphs connecting visitors to each other
2. **Contextual Warnings in Documentation**
 - This document explicitly labels privacy risks
 - Any published analysis will include disclaimers about re-identification
3. **Minimization Principle**
 - Only extracted necessary fields for analysis goals
 - Did not scrape additional public data to “enrich” records

Recommended for Future Use:

1. **K-Anonymity for Public Visualizations**
 - Suppress visitor names when count < 5 visits
 - Aggregate low-frequency visitees into “Other” category
2. **Temporal Aggregation**
 - Report monthly totals rather than daily visit logs
 - Reduces granularity for pattern tracking
3. **Differential Privacy Mechanisms**
 - Add statistical noise to aggregate counts
 - Prevents exact inference from subtraction attacks

1.9.2 7.2 Policy Recommendations

For Data Publishers (White House):

1. **Standardize Data Formats**
 - Consistent column naming across releases
 - Publish data dictionary with each release
 - Provide technical documentation of redaction policies

2. **Risk Assessment Transparency**
 - Publish Privacy Impact Assessment (PIA) for WAVES releases
 - Explain what data is redacted and why
 - Provide notice of policy changes
3. **Visitor Notice Enhancement**
 - Ensure ALL visitors understand WAVES policy
 - Provide opt-out for non-official visitors (e.g., tour participants)
 - Post clear signage at all entry points

For Data Users (Researchers, Journalists):

1. **Responsible Use Guidelines**
 - Do not publish individual-level data without newsworthiness justification
 - Redact names when analyzing patterns (use IDs instead)
 - Consider harm potential before publishing findings
2. **Security Protocols**
 - Store data securely (encrypted, access-controlled)
 - Do not cross-reference with other datasets without IRB approval
 - Delete data after analysis completion
3. **Transparency in Methods**
 - Document all transformations (as done in this analysis)
 - Publish code and methodology for reproducibility
 - Clearly state assumptions and limitations

1.9.3 7.3 Stakeholder Engagement

Public Interest Advocacy:

- **Who:** Civil liberties organizations (ACLU, EFF)
- **Action:** Monitor WAVES releases for privacy violations
- **Goal:** Balance transparency with individual rights

Academic Oversight:

- **Who:** University Institutional Review Boards (IRBs)
- **Action:** Treat public data with same ethical rigor as private data
- **Goal:** Establish research ethics standards for government data

Legislative Action:

- **Who:** Congressional oversight committees
- **Action:** Codify WAVES release requirements and privacy protections
- **Goal:** Make voluntary disclosure mandatory and standardized

1.10 8. Key Findings and Insights

1.10.1 8.1 Transparency Achievements

Over 1.2 million records made public demonstrates commitment to accountability
Systematic release schedule enables longitudinal analysis of access patterns
Granular detail allows verification of individual claims about White House access

1.10.2 8.2 Privacy and Security Concerns

Personal information of ~100,000 unique individuals is indefinitely public

Pattern analysis could reveal sensitive government operations or vulnerabilities

Dual-use data serves democracy but also enables harassment/tracking

1.10.3 8.3 Data Quality Issues

Column standardization failure creates analysis barriers

Missing data (70-90% in some fields) limits analytical utility

Documentation gaps leave analysts making assumptions about data meaning

1.11 9. Call to Action: Ethical Data Stewardship Framework

1.11.1 For Government Agencies:

1. **Adopt “Privacy by Design”** - Build privacy protections into data systems from inception
2. **Publish Data Dictionaries** - Every dataset needs comprehensive documentation
3. **Conduct PIAs** - Make privacy impact assessments public and mandatory

1.11.2 For Data Analysts:

1. **Ask “Should I?”** before “Can I?” - Just because data is public doesn’t mean all uses are ethical
2. **Document Everything** - Your transformation decisions have ethical implications
3. **Minimize Exposure** - Aggregate, anonymize, and contextualize before publishing

1.11.3 For Citizens:

1. **Demand Transparency** - Government accountability requires accessible data
2. **Protect Privacy** - Support laws limiting re-identification and secondary use
3. **Stay Informed** - Understand what information about you is public

1.11.4 For Policymakers:

1. **Mandate Standards** - Require consistent data formats across agencies
2. **Balance Access** - Create legal frameworks that protect both transparency and privacy
3. **Fund Oversight** - Support agencies that audit data release practices

1.12 10. Conclusion

The White House WAVES dataset represents a **grand experiment in democratic transparency**. It proves that open government data is feasible at scale—but it also reveals the **ethical tensions** inherent in making personal information public.

This analysis demonstrates that **ethical data practice is not just about legal compliance**. It requires: - Transparent documentation of decisions - Honest assessment of risks - Proactive mitigation of harms - Ongoing dialogue with stakeholders

The 1.2 million records analyzed here are more than data points—they represent **real people** who visited their government. Our obligation is to use this data in ways that **strengthen democracy** without **compromising individual dignity**.

The question is not whether to release government data, but how to do so responsibly.

1.13 Appendix: Technical Metadata

Analysis Date: January 3, 2026

Dataset: WAVES_Combined_2021-2023.csv

Total Records: 1,207,458

Date Range: 2021-01-01 to 2023-12-31

Source Files: 25 CSV files from WhiteHouse.gov WAVES releases

Analyst: [Configured for educational exercise]

Tools Used: Python 3.13.5, Pandas 2.x

Repository: Local analysis, not published

Contact for Questions: [Ethics review recommended before public release]

Document Version: 1.0

Last Updated: January 3, 2026

License: Analysis methodology freely reusable; source data subject to government terms of use

[]: